

◆ Chunk 1: What is Spring Boot

Title: What is Spring Boot

Definition:

Spring Boot is an open-source Java framework built on top of the Spring Framework that simplifies backend development using auto-configuration and embedded servers.

Key Features:

- Auto-configuration
- Embedded server (Tomcat)
- Starter dependencies
- Minimal configuration

Use Case:

Used to build REST APIs and microservices quickly.

◆ Chunk 2: Spring Boot Advantages

Title: Advantages of Spring Boot

Benefits:

- Faster development
 - Less boilerplate code
 - Easy deployment (JAR file)
 - Built-in production tools
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◆ Chunk 3: Project Structure

Title: Spring Boot Project Structure

Components:

- Controller → handles HTTP
- Service → business logic
- Repository → database
- Model → entity
- DTO → data transfer

◆ Chunk 4: Main Class

Title: Spring Boot Main Class

Explanation:

Entry point of application.

Example:

```
@SpringBootApplication
public class App {
    public static void main(String[] args) {
        SpringApplication.run(App.class, args);
    }
}
```

◆ Chunk 5: Dependency Injection

Title: Dependency Injection

Definition:

Spring injects dependencies automatically instead of manual creation.

Benefits:

- Loose coupling
- Easier testing

◆ Chunk 6: @RestController

Title: @RestController

Definition:

Used to create REST APIs returning JSON responses.

◆ Chunk 7: HTTP Methods

Title: REST HTTP Methods

Methods:

- GET → read
- POST → create

- PUT → update
 - DELETE → delete
-

◆ Chunk 8: JPA

Title: What is JPA

Definition:

JPA is a Java specification for database operations.

Implementation:

Hibernate

◆ Chunk 9: Repository Layer

Title: Spring Data Repository

Definition:

Handles database operations.

Example:

```
public interface UserRepo extends JpaRepository<User, Long> {}
```

◆ Chunk 10: Service Layer

Title: Service Layer

Role:

Contains business logic.

◆ Chunk 11: Controller Layer

Title: Controller Layer

Role:

Handles HTTP requests and responses.

◆ Chunk 12: DTO Pattern

Title: DTO Pattern

Definition:

Used to transfer data between layers.

Benefits:

- Security
 - Clean API
-

◆ Chunk 13: Validation

Title: Validation in Spring Boot

Definition:

Validates user input using annotations.

Examples:

- @NotNull
 - @Email
 - @Size
-

◆ Chunk 14: Exception Handling

Title: Global Exception Handling

Definition:

Handles errors using @ControllerAdvice.

◆ Chunk 15: Spring Security

Title: Spring Security

Definition:

Framework for authentication and authorization.

◆ Chunk 16: JWT

Title: JWT Authentication

Definition:

Token-based authentication.

Structure:

Header.Payload.Signature

◆ Chunk 17: JWT Flow

Title: JWT Flow

Steps:

1. User login
 2. Server generates token
 3. Client sends token
 4. Server validates
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◆ Chunk 18: Password Encoding

Title: Password Encoding

Definition:

Passwords are stored securely using hashing.

Example:

BCrypt

◆ Chunk 19: Pagination

Title: Pagination

Definition:

Fetching data in pages.

Tools:

Pageable, PageRequest

◆ Chunk 20: Sorting

Title: Sorting

Definition:

Sorting results using fields.

◆ Chunk 21: File Upload

Title: File Upload

Definition:

Uploading files using MultipartFile.

◆ Chunk 22: Logging

Title: Logging

Definition:

Tracking application events.

Tool:

SLF4J

◆ Chunk 23: Testing

Title: Testing

Types:

- Unit test
- Integration test

Tools:

JUnit, Mockito

◆ Chunk 24: @Transactional

Title: @Transactional

Definition:

Ensures database operations are atomic.

◆ Chunk 25: Relationships

Title: JPA Relationships

Types:

- OneToOne
 - OneToMany
 - ManyToMany
-

◆ Chunk 26: Fetch Types

Title: Fetch Types

Types:

- LAZY
 - EAGER
-

◆ Chunk 27: Profiles

Title: Spring Profiles

Definition:

Different configs for dev, prod.

◆ Chunk 28: Application Properties

Title: application.properties

Purpose:

Stores configuration like DB, ports.

◆ Chunk 29: Actuator

Title: Spring Boot Actuator

Definition:

Provides health and metrics endpoints.

◆ Chunk 30: Best Practices

Title: Spring Boot Best Practices

Guidelines:

- Use DTO
- Use constructor injection
- Keep controllers thin
- Use global exception handling